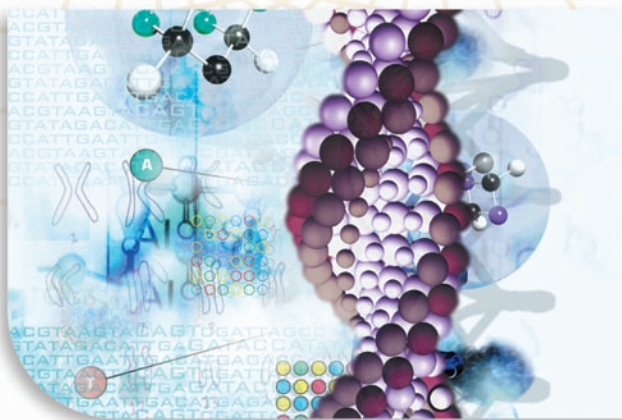


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Preface

The twentieth century has been called “the genetic century,” and rightly so: The genetic revolution began with the rediscovery of Gregor Mendel’s work in 1900, Watson and Crick elucidated the structure of DNA in 1953, and the first draft of the human genome sequence was announced in February 2001. As dramatic and important as these advances are, however, they will almost certainly pale when compared to those still awaiting us. Building on foundations laid over the last one hundred years, the twenty-first century will likely see discoveries that profoundly affect our understanding of our genetic nature, and greatly increase our ability to manipulate genes to shape ourselves and our environment. As more is learned, the pace of discovery will only increase, revealing not only the identities of increasing numbers of genes, but more importantly, how they function, interact, and, in some cases, cause disease.

As the importance of genetics in our daily lives has grown, so too has the importance of its place in the modern science classroom: In the study of biology, genetics has become the central science. Our purpose in creating this encyclopedia is to provide students and teachers the most comprehensive and accessible reference available for understanding this rapidly changing field.

A Comprehensive Reference

In the four volumes of *Genetics*, students will find detailed coverage of every topic included in standard and advanced biology courses, from fundamental concepts to cutting-edge applications, as well as topics so new that they have not yet become a part of the regular curriculum. The set explores the history, theory, technology, and uses (and misuses) of genetic knowledge. Topics span the field from “classical” genetics to molecular genetics to population genetics. Students and teachers can use the set to reinforce classroom lessons about basic genetic processes, to expand on a discussion of a special topic, or to learn about an entirely new idea.

Genetic Disorders and Social Issues

Many advances in genetics have had their greatest impact on our understanding of human health and disease. One of the most important areas of research is in the understanding of complex diseases, such as cancer and Alzheimer’s disease, in which genes and environment interact to produce or prevent disease. *Genetics* devotes more than two dozen entries to both single-gene and complex genetic disorders, offering the latest understanding of

★Explore further in Gene, Polymerase Chain Reaction, and Eugenics



★Explore further in
Alzheimer Disease,
Genetic Testing, and
Gene Therapy

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Cloning: Ethical Issues

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Sequencing DNA, DNA
Microarrays, and Internet

★Explore further in
Morgan, Thomas Hunt,
and Computational
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their causes, diagnoses, and treatments. Many more entries illustrate basic genetic processes with discussion of the diseases in which these processes go wrong. In addition, students will find in-depth explanations of how genetic diseases arise, how disease genes are discovered, and how gene therapy hopes to treat them.

Advances in our understanding of genetics and improvements in techniques of genetic manipulation have brought great benefits, but have also raised troubling ethical and legal issues, most prominently in the areas of reproductive technology, cloning, and biotechnology. In *Genetics*, students will find discussions of both the science behind these advances and the ethical issues each has engendered. As with nearly every entry in *Genetics*, these articles are accompanied by suggestions for further reading to allow the student to seek more depth and pursue other points of view.

The Tools of the Trade

The explosion of genetic knowledge in the last several decades can be attributed in large part to the discovery and development of a set of precise and powerful tools for analyzing and manipulating DNA. In these volumes, students will find clear explanations of how each of these tools work, as well as how they are used by scientists to conduct molecular genetic research. We also discuss how the computer and the Internet have radically expanded the ability of scientists to process large amounts of data. These technologies have made it possible to analyze whole genomes, leading not just to the discovery of new genes, but to a greater understanding of how entire genomes function and evolve.

The Past and the Future

The short history of genetics is marked by brilliant insights and major theoretical advances, as well as misunderstandings and missed opportunities. *Genetics* examines these events in both historical essays and biographies of major figures, from Mendel to McClintock. The future of genetics will be created by today's students, and in these volumes we present information on almost two dozen careers in this field, ranging from attorney to clinical geneticist to computational biologist.

Contributors and Arrangement of the Material

The goal of each of the 253 entries in *Genetics* is to give the interested student access to a depth of discussion not easily available elsewhere. Entries have been written by professionals in the field of genetics, including experts whose work has helped define the current state of knowledge. All of the entries have been written with the needs of students in mind, and they all provide the background and context necessary to help students make connections with classroom lessons.

To aid understanding and increase interest, most entries are illustrated with clear diagrams and dramatic photographs. Each entry is followed by cross-references to related entries, and most have a list of suggested readings and/or Internet resources for further exploration or elaboration. Specialized or unfamiliar terms are defined in the margin and collected in a glossary at the end of each volume. Each volume also contains an index, and

a cumulative index is found at the end of volume four. A topical index is also included, allowing students and teachers to see at a glance the range of entries available on a particular topic.

Acknowledgments and Thanks

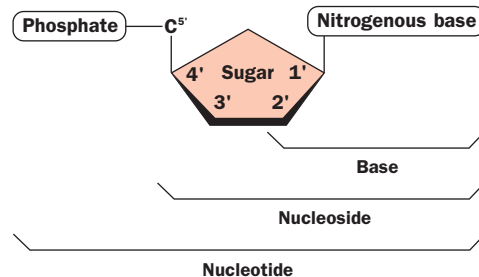
Genetics represents the collective inspiration and hard work of many people. Hélène Potter at Macmillan Reference USA knew how important a reference this encyclopedia would be, and her commitment and enthusiasm brought it into being. Kate Millson has provided simply outstanding editorial management throughout this long process, and I am deeply in her debt. Our three editorial board members—Ralph R. Meyer, David A. Micklos, and Margaret A. Pericak-Vance—gave the encyclopedia its broad scope and currency, and were vital in ensuring accuracy in this rapidly changing field. Finally, the entries in *Genetics* are the product of well over one hundred scientists, doctors, and other professionals. Their willingness to contribute their time and expertise made this work possible, and it is to them that the greatest thanks are due.

Richard Robinson
Tucson, Arizona
rrobinson@nasw.org

For Your Reference

The following section provides a group of diagrams and illustrations applicable to many entries in this encyclopedia. The molecular structures of DNA and RNA are provided in detail in several different formats, to help the student understand the structures and visualize how these molecules combine and interact. The full set of human chromosomes are presented diagrammatically, each of which is shown with a representative few of the hundreds or thousands of genes it carries.

NUCLEOTIDE STRUCTURE



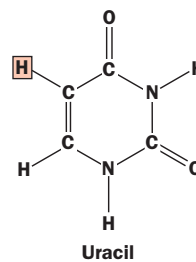
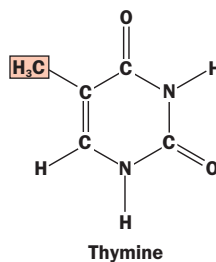
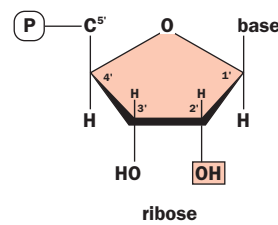
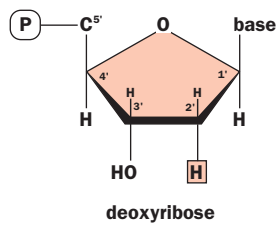
Sample naming conventions for each structure:

Adenine

Adenosine

Adenosine monophosphate

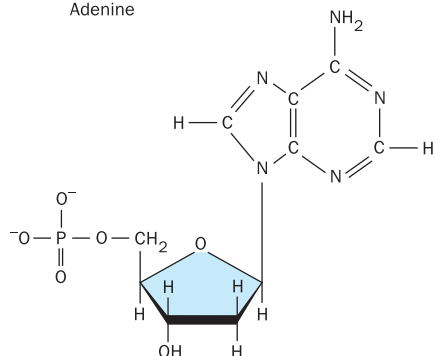
DNA VS. RNA



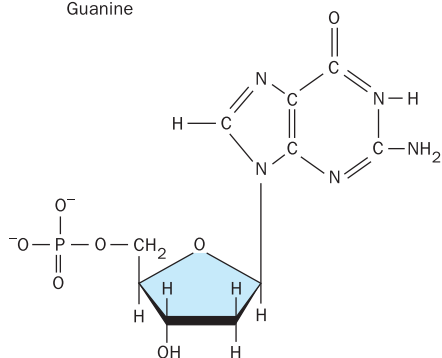
NUCLEOTIDE STRUCTURES

Purine-containing DNA nucleotides

Adenine

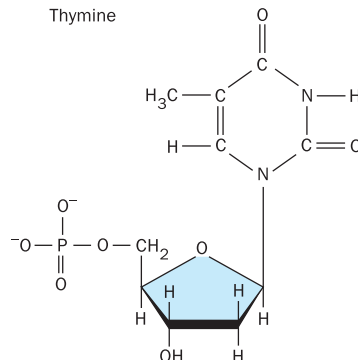


Guanine

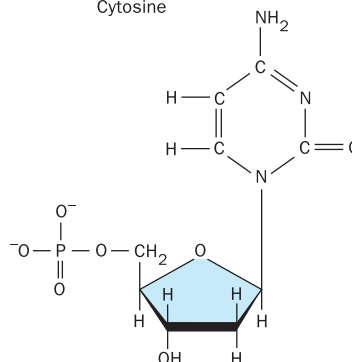


Pyrimidine-containing DNA nucleotides

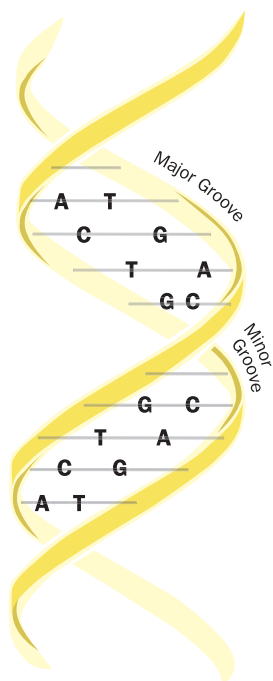
Thymine



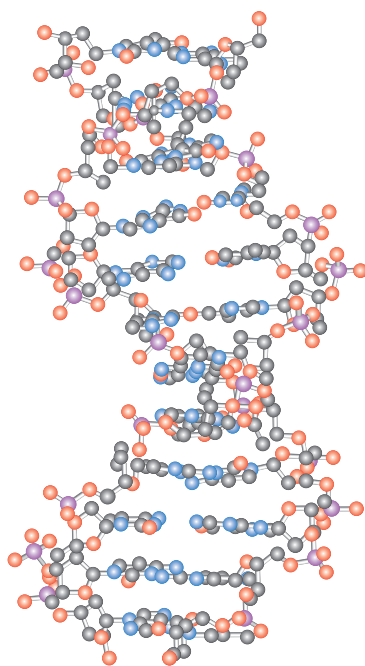
Cytosine



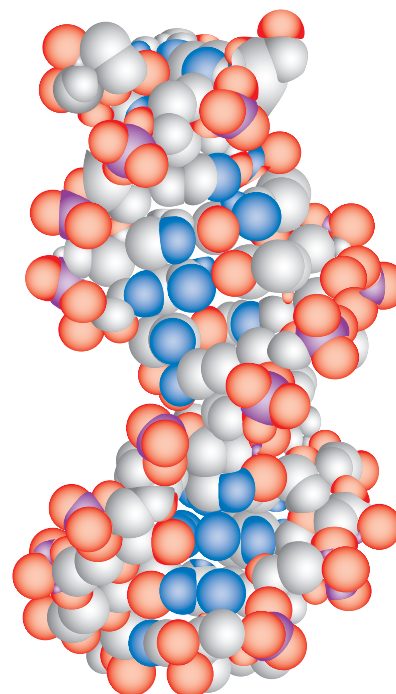
CANONICAL B-DNA DOUBLE HELIX



Ribbon model

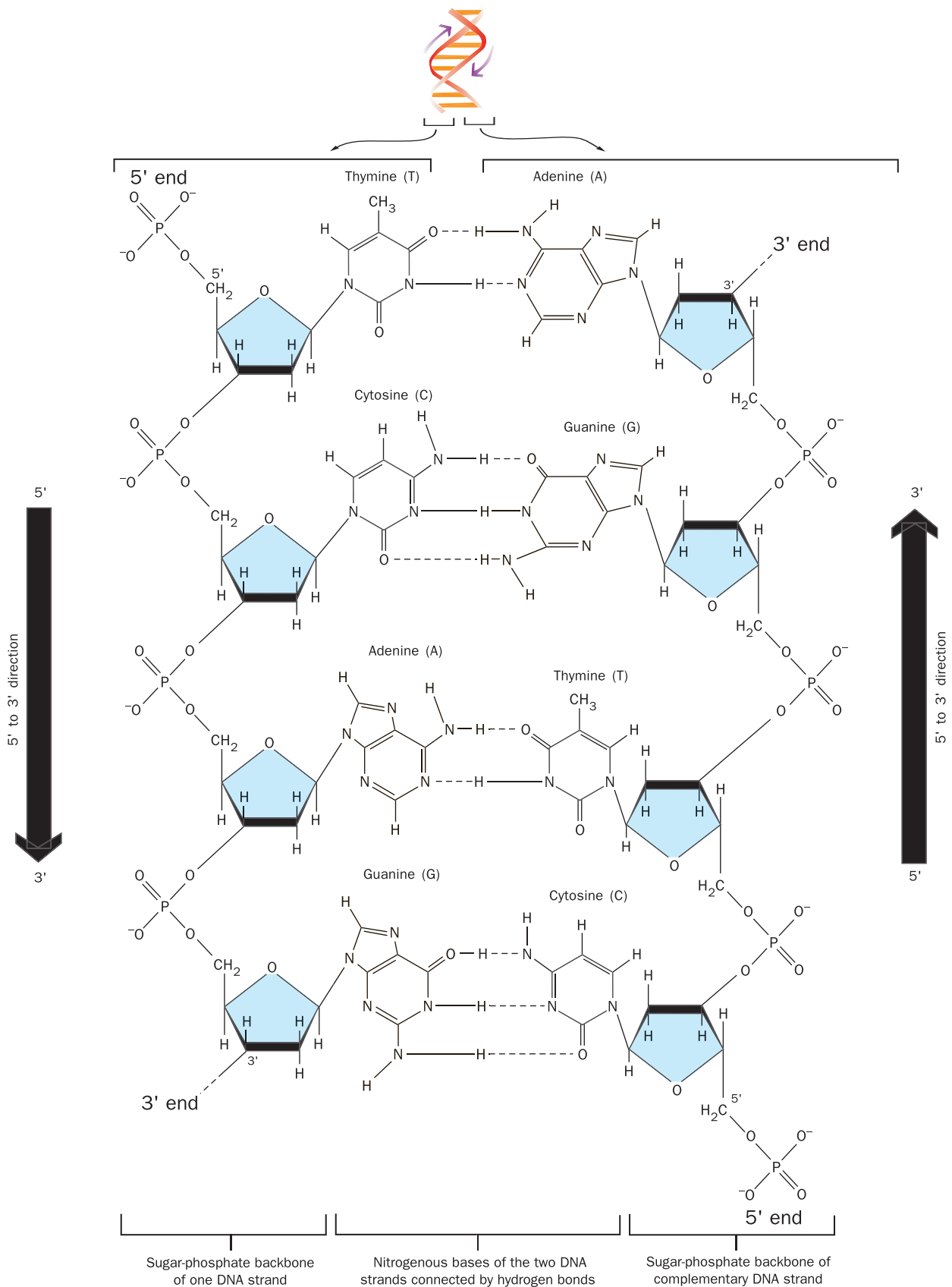


Ball-and-stick model

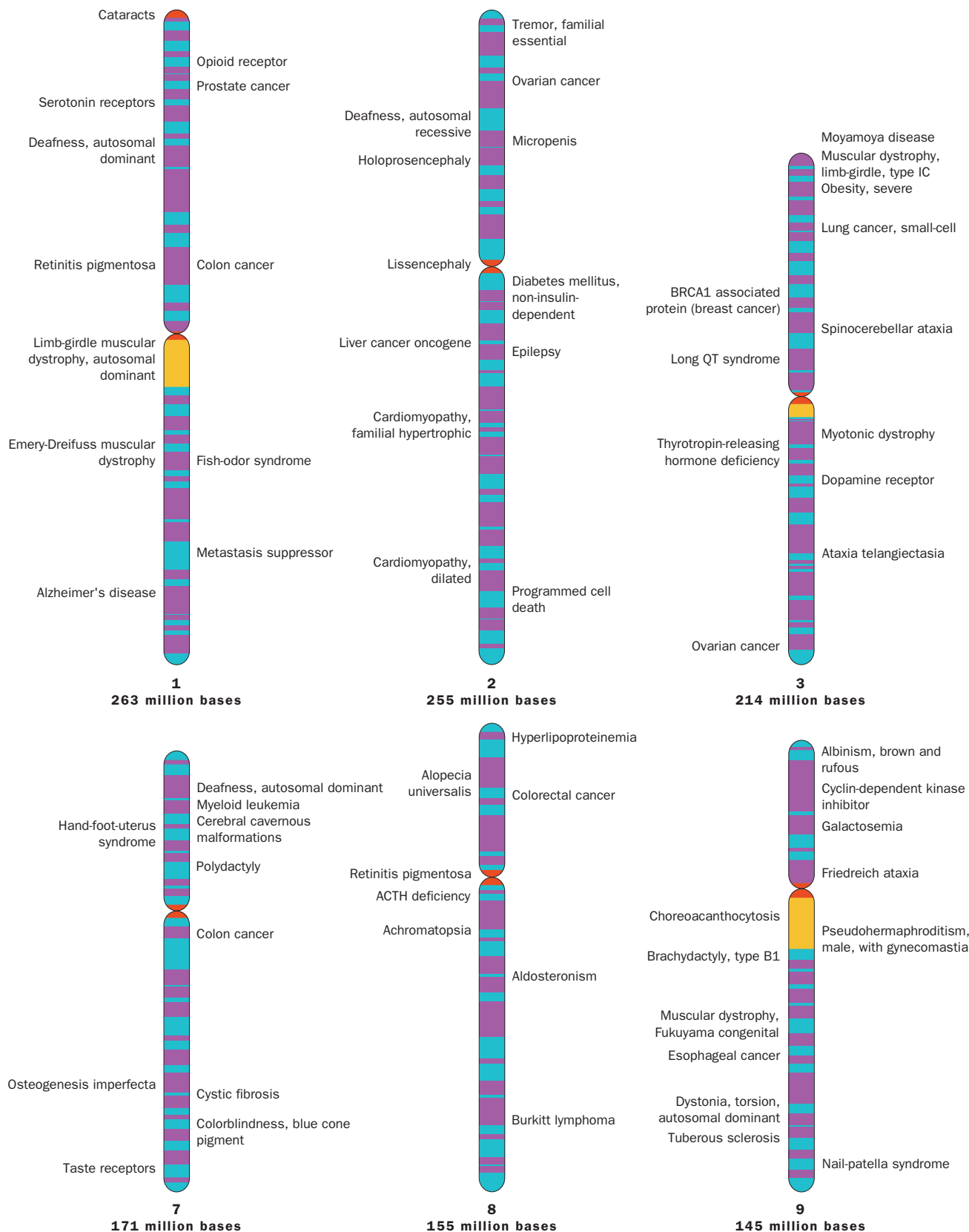


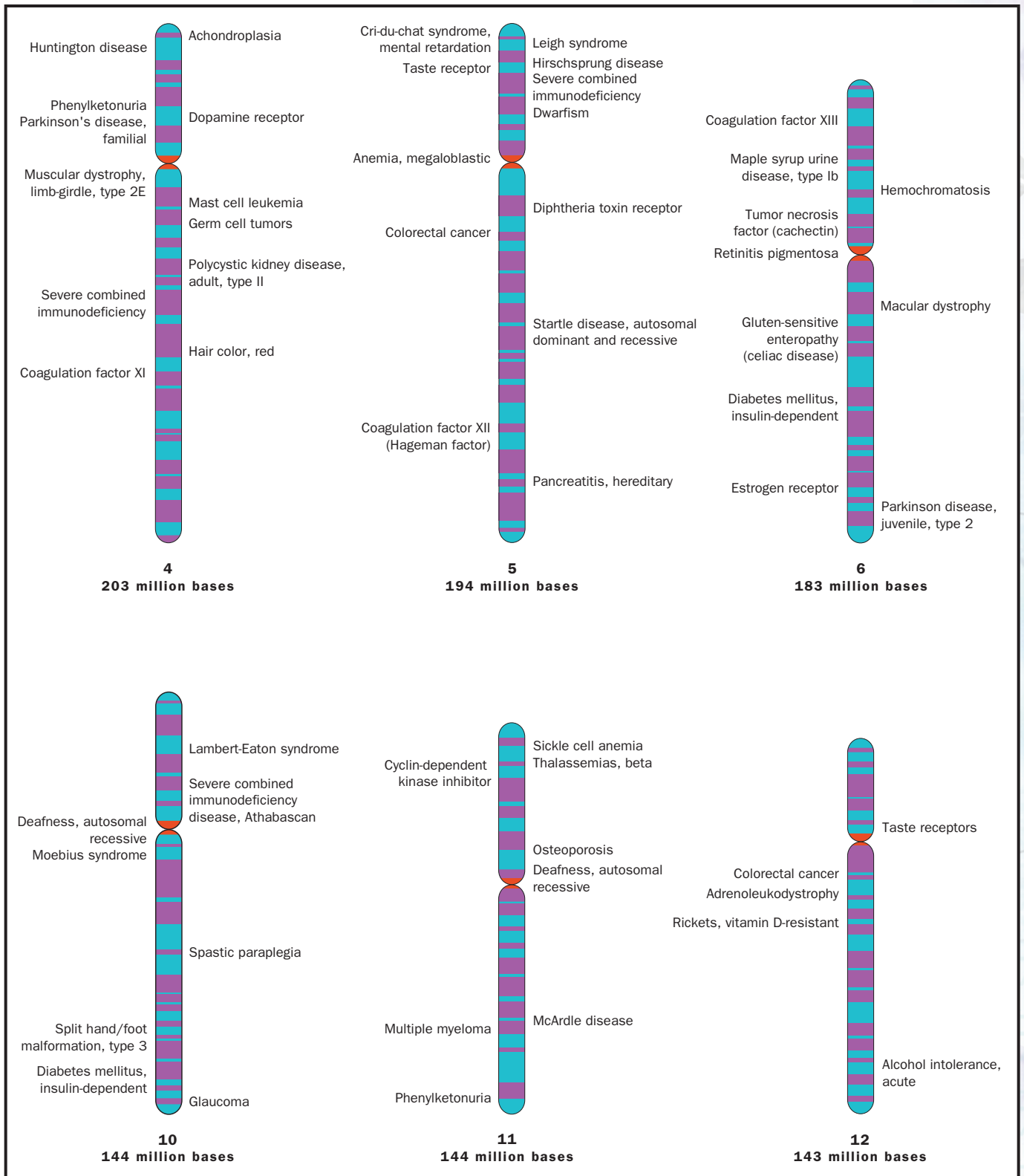
Space-filling model

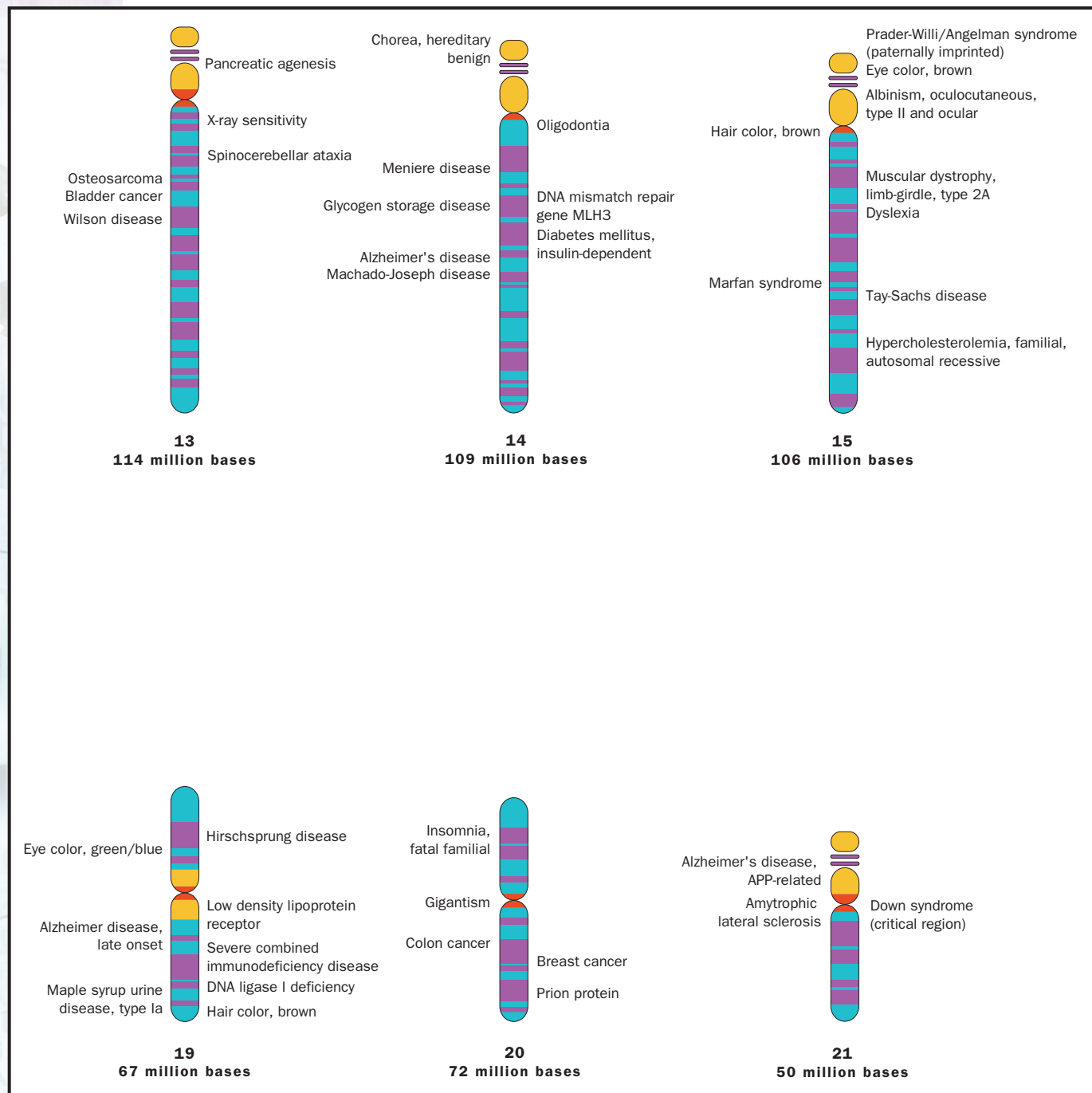
DNA NUCLEOTIDES PAIR UP ACROSS THE DOUBLE HELIX; THE TWO STRANDS RUN ANTI-PARALLEL

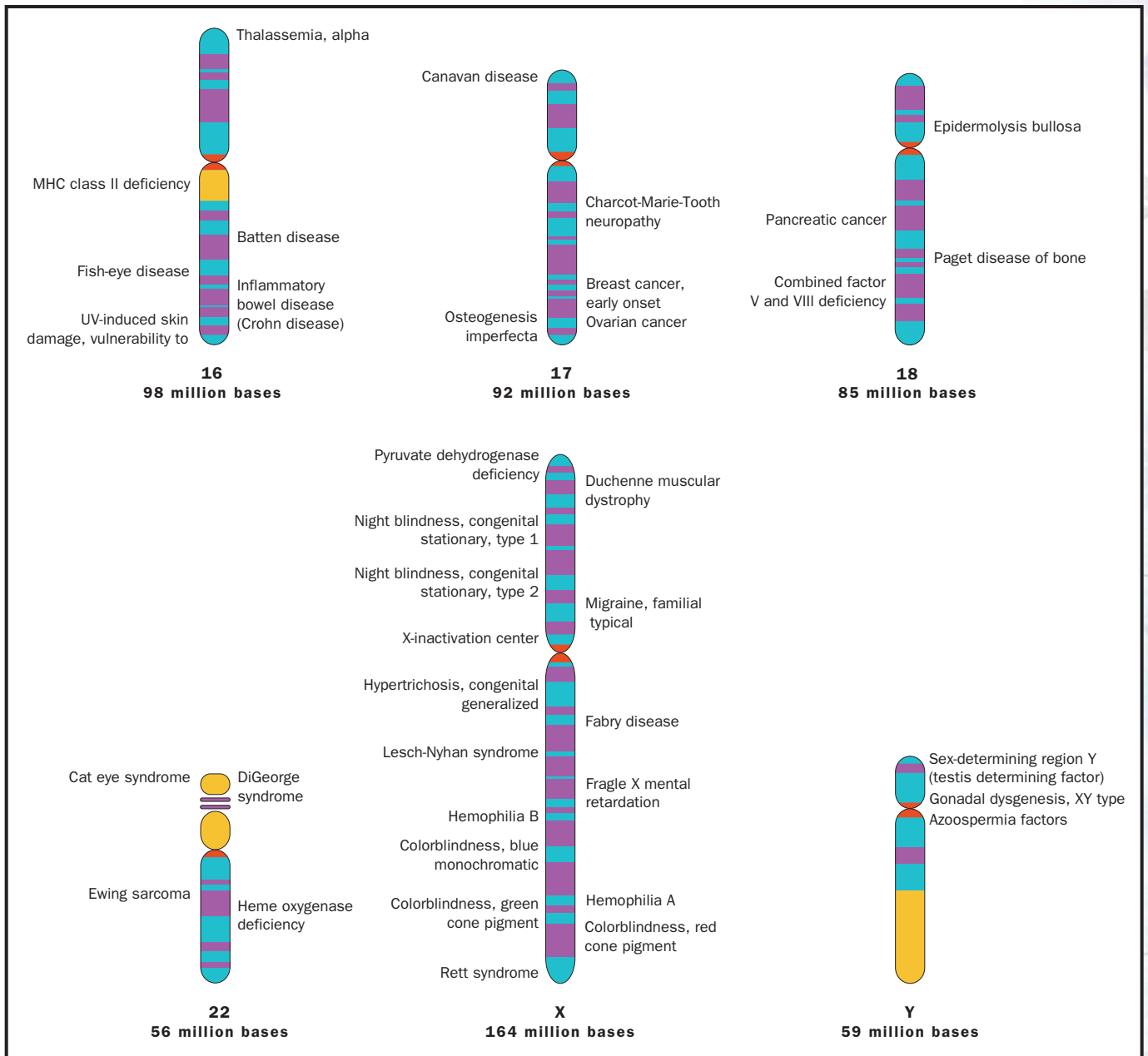


SELECTED LANDMARKS OF THE HUMAN GENOME









Contributors

- Eric Aamodt
Louisiana State University Health Sciences Center, Shreveport
Gene Expression: Overview of Control
- Maria Cristina Abilock
Applied Biosystems
Automated Sequencer
Cycle Sequencing
Protein Sequencing
Sequencing DNA
- Ruth Abramson
University of South Carolina School of Medicine
Intelligence
Psychiatric Disorders
Sexual Orientation
- Stanley Ambrose
University of Illinois
Population Bottleneck
- Allison Ashley-Koch
Duke Center for Human Genetics
Disease, Genetics of
Fragile X Syndrome
Geneticist
- David T. Auble
University of Virginia Health System
Transcription
- Bruce Barshop
University of California, San Diego
Metabolic Disease
- Mark A. Batzer
Louisiana State University
Pseudogenes
Repetitive DNA Elements
Transposable Genetic Elements
- Robert C. Baumiller
Xavier University
Reproductive Technology
Reproductive Technology: Ethical Issues
- Mary Beckman
Idaho Falls, Idaho
DNA Profiling
HIV
- Samuel E. Bennett
Oregon State University
Department of Genetics
DNA Repair
Laboratory Technician
Molecular Biologist
- Andrea Bernasconi
Cambridge University, U.K.
Multiple Alleles
Nondisjunction
- C. William Birky, Jr.
University of Arizona
Inheritance, Extranuclear
- Joanna Bloom
New York University Medical Center
Cell Cycle
- Deborah Blum
University of Wisconsin, Madison
Science Writer
- Bruce Blumberg
University of California, Irvine
Hormonal Regulation
- Suzanne Bradshaw
University of Cincinnati
Transgenic Animals
Yeast
- Carolyn J. Brown
University of British Columbia
Mosaicism
- Michael J. Bumbulis
Baldwin-Wallace College
Blotting
- Michael Buratovich
Spring Arbor College
Operon
- Elof Carlson
The State University of New York, Stony Brook
Chromosomal Theory of Inheritance, History
Gene
Muller, Hermann
Polyploidy
Selection
- Regina Carney
Duke University
College Professor
- Shu G. Chen
Case Western Reserve University
Prion
- Gwen V. Childs
University of Arkansas for Medical Sciences
In situ Hybridization
- Cindy T. Christen
Iowa State University
Technical Writer
- Patricia L. Clark
University of Notre Dame
Chaperones
- Steven S. Clark
University of Wisconsin
Oncogenes
- Nathaniel Comfort
George Washington University
McClintock, Barbara
- P. Michael Conneally
Indiana University School of Medicine
Blood Type
Epistasis
Heterozygote Advantage
- Howard Cooke
Western General Hospital: MRC Human Genetics Unit
Chromosomes, Artificial
- Denise E. Costich
Boyce Thompson Institute
Maize
- Terri Creeden
March of Dimes
Birth Defects
- Kenneth W. Culver
Novartis Pharmaceuticals Corporation
Genomics
Genomics Industry
Pharmaceutical Scientist
- Mary B. Daly
Fox Chase Cancer Center
Breast Cancer
- Pieter de Haseth
Case Western Reserve University
Transcription



- Rob DeSalle
American Museum of Natural History
Conservation Geneticist
Conservation Biology: Genetic Approaches
- Elizabeth A. De Stasio
Lawrence University
Cloning Organisms
- Danielle M. Dick
Indiana University
Behavior
- Michael Dietrich
Dartmouth College
Nature of the Gene, History
- Christine M. Disteche
University of Washington
X Chromosome
- Gregory Michael Dorr
University of Alabama
Eugenics
- Jennie Dusheck
Santa Cruz, California
Population Genetics
- Susanne D. Dyby
U.S. Department of Agriculture: Center for Medical, Agricultural, and Veterinary Entomology
Classical Hybrid Genetics
Mendelian Genetics
Pleiotropy
- Barbara Emberson Soots
Folsom, California
Agricultural Biotechnology
- Susan E. Estabrooks
Duke Center for Human Genetics
Fertilization
Genetic Counselor
Genetic Testing
- Stephen V. Faraone
Harvard Medical School
Attention Deficit Hyperactivity Disorder
- Gerald L. Feldman
Wayne State University Center for Molecular Medicine and Genetics
Down Syndrome
- Linnea Fletcher
Bio-Link South Central Regional Coordinator, Austin Community College
Educator
Gel Electrophoresis
Marker Systems
Plasmid
- Michael Fossel
Executive Director, American Aging Association
Accelerated Aging: Progeria
- Carol L. Freund
National Institute of Health: Warren G. Magnuson Clinical Center
Genetic Testing: Ethical Issues
- Joseph G. Gall
Carnegie Institution
Centromere
- Darrell R. Galloway
The Ohio State University
DNA Vaccines
- Pierluigi Gambetti
Case Western Reserve University
Prion
- Robert F. Garry
Tulane University School of Medicine
Retrovirus
Virus
- Perry Craig Gaskell, Jr.
Duke Center for Human Genetics
Alzheimer's Disease
- Theresa Geiman
National Institute of Health: Laboratory of Receptor Biology and Gene Expression
Methylation
- Seth G. N. Grant
University of Edinburgh
Embryonic Stem Cells
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University of Calgary
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Cystic Fibrosis
- Charles J. Grossman
Xavier University
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- Cynthia Guidi
University of Massachusetts Medical School
Chromosome, Eukaryotic
- Patrick G. Guilfoile
Bemidji State University
DNA Footprinting
Microbiologist
Recombinant DNA
Restriction Enzymes
- Richard Haas
University of California Medical Center
Mitochondrial Diseases
- William J. Hagan
College of St. Rose
Evolution, Molecular
- Jonathan L. Haines
Vanderbilt University Medical Center
Complex Traits
Human Disease Genes, Identification of
- Mapping
McKusick, Victor
- Michael A. Hauser
Duke Center for Human Genetics
DNA Microarrays
Gene Therapy
- Leonard Hayflick
University of California
Telomere
- Shaun Heaphy
University of Leicester, U.K.
Viroids and Virusoids
- John Heddle
York University
Mutagenesis
Mutation
Mutation Rate
- William Horton
Sriners Hospital for Children
Growth Disorders
- Brian Hoyle
Square Rainbow Limited
Overlapping Genes
- Anthony N. Imbalzano
University of Massachusetts Medical School
Chromosome, Eukaryotic
- Nandita Jha
University of California, Los Angeles
Triplet Repeat Disease
- John R. Jungck
Beloit College
Gene Families
- Richard Karp
Department of Biological Sciences, University of Cincinnati
Transplantation
- David H. Kass
Eastern Michigan University
Pseudogenes
Transposable Genetic Elements
- Michael L. Kochman
University of Pennsylvania Cancer Center
Colon Cancer
- Bill Kraus
Duke University Medical Center
Cardiovascular Disease
- Steven Krawiec
Lehigh University
Genome
- Mark A. Labow
Novartis Pharmaceuticals Corporation
Genomics
Genomics Industry
Pharmaceutical Scientist
- Ricki Lewis
McGraw-Hill Higher Education; The Scientist
Bioremediation
Biotechnology: Ethical Issues
Cloning: Ethical Issues



- Genetically Modified Foods
Plant Genetic Engineer
Prenatal Diagnosis
Transgenic Organisms: Ethical Issues
Lasse Lindahl
University of Maryland, Baltimore
Ribozyme
RNA
David E. Loren
University of Pennsylvania School of Medicine
Colon Cancer
Dennis N. Luck
Oberlin College
Biotechnology
Jeanne M. Lusher
Wayne State University School of Medicine; Children's Hospital of Michigan
Hemophilia
Kamrin T. MacKnight
Medlen, Carroll, LLP: Patent, Trademark and Copyright Attorneys
Attorney
Legal Issues
Patenting Genes
Privacy
Jarema Malicki
Harvard Medical School
Zebrafish
Eden R. Martin
Duke Center for Human Genetics
Founder Effect
Inbreeding
William Mattox
University of Texas/Anderson Cancer Center
Sex Determination
Brent McCown
University of Wisconsin
Transgenic Plants
Elizabeth C. Melvin
Duke Center for Human Genetics
Gene Therapy: Ethical Issues
Pedigree
Ralph R. Meyer
University of Cincinnati
Biotechnology and Genetic Engineering, History of
Chromosome, Eukaryotic
Genetic Code
Human Genome Project
Kenneth V. Mills
College of the Holy Cross
Post-translational Control
Jason H. Moore
Vanderbilt University Medical School
Quantitative Traits
Statistical Geneticist
Statistics
Dale Mosbaugh
Oregon State University: Center for Gene Research and Biotechnology
DNA Repair
Laboratory Technician
Molecular Biologist
Paul J. Muhlrad
University of Arizona
Alternative Splicing
Apoptosis
Arabidopsis thaliana
Cloning Genes
Combinatorial Chemistry
Fruit Fly: *Drosophila*
Internet
Model Organisms
Pharmacogenetics and Pharmacogenomics
Polymerase Chain Reaction
Cynthia A. Needham
Boston University School of Medicine
Archaea
Conjugation
Transgenic Microorganisms
R. John Nelson
University of Victoria
Balanced Polymorphism
Gene Flow
Genetic Drift
Polymorphisms
Speciation
Carol S. Newlon
University of Medicine and Dentistry of New Jersey
Replication
Sophia A. Oliveria
Duke University Center for Human Genetics
Gene Discovery
Richard A. Padgett
Lerner Research Institute
RNA Processing
Michele Pagano
New York University Medical Center
Cell Cycle
Rebecca Pearlman
Johns Hopkins University
Probability
Fred W. Perrino
Wake Forest University School of Medicine
DNA Polymerases
Nucleases
Nucleotide
David Pimentel
Cornell University: College of Agriculture and Life Sciences
Biopesticides
Toni I. Pollin
University of Maryland School of Medicine
Diabetes
Sandra G. Porter
Geospiza, Inc.
Homology
Eric A. Postel
Duke University Medical Center
Color Vision
Eye Color
Prema Rapuri
Creighton University
HPLC: High-Performance Liquid Chromatography
Anthony J. Recupero
Gene Logic
Bioinformatics
Biotechnology Entrepreneur
Proteomics
Diane C. Rein
BioComm Consultants
Clinical Geneticist
Nucleus
Roundworm: *Caenorhabditis elegans*
Severe Combined Immune Deficiency
Jacqueline Bebout Rimmner
Duke Center for Human Genetics
Chromosomal Aberrations
Keith Robertson
Epigenetic Gene Regulation and Cancer Institute
Methylation
Richard Robinson
Tucson, Arizona
Androgen Insensitivity Syndrome
Antisense Nucleotides
Cell, Eukaryotic
Crick, Francis
Delbrück, Max
Development, Genetic Control of
DNA Structure and Function, History
Eubacteria
Evolution of Genes
Hardy-Weinberg Equilibrium
High-Throughput Screening
Immune System Genetics
Imprinting
Inheritance Patterns
Mass Spectrometry
Mendel, Gregor
Molecular Anthropology
Morgan, Thomas Hunt
Mutagen
Purification of DNA
RNA Interference
RNA Polymerases
Transcription Factors
Twins
Watson, James
Richard J. Rose
Indiana University
Behavior
Howard C. Rosenbaum
Science Resource Center, Wildlife Conservation Society
Conservation Geneticist
Conservation Biology: Genetic Approaches

Astrid M. Roy-Engel
*Tulane University Health Sciences
Center*
Repetitive DNA Elements

Joellen M. Schildkraut
Duke University Medical Center
Public Health, Genetic Tech-
niques in

Silke Schmidt
Duke Center for Human Genetics
Meiosis
Mitosis

David A. Scicchitano
New York University
Ames Test
Carcinogens

William K. Scott
Duke Center for Human Genetics
Aging and Life Span
Epidemiologist
Gene and Environment

Gerry Shaw
*MacKnight Brain Institute of the
University of Florida*
Signal Transduction

Alan R. Shuldiner
*University of Maryland School of
Medicine*
Diabetes

Richard R. Sinden
*Institute for Biosciences and
Technology: Center for Genome
Research*
DNA

Paul K. Small
Eureka College
Antibiotic Resistance
Proteins
Reading Frame

Marcy C. Speer
Duke Center for Human Genetics
Crossing Over
Founder Effect
Inbreeding
Individual Genetic Variation
Linkage and Recombination

Jeffrey M. Stajich
Duke Center for Human Genetics
Muscular Dystrophy

Judith E. Stenger
Duke Center for Human Genetics
Computational Biologist
Information Systems Manager

Frank H. Stephenson
Applied Biosystems
Automated Sequencer
Cycle Sequencing
Protein Sequencing
Sequencing DNA

Gregory Stewart
State University of West Georgia
Transduction
Transformation

Douglas J. C. Strathdee
University of Edinburgh
Embryonic Stem Cells
Gene Targeting
Rodent Models

Jeremy Sugarman
*Duke University Department of
Medicine*
Genetic Testing: Ethical Issues

Caroline M. Tanner
Parkinson's Institute
Twins

Alice Telesnitsky
University of Michigan
Reverse Transcriptase

Daniel J. Tomso
*National Institute of Environmental
Health Sciences*
DNA Libraries
Escherichia coli
Genetics

Angela Trepanier
*Wayne State University Genetic
Counseling Graduate Program*
Down Syndrome

Peter A. Underhill
Stanford University
Y Chromosome

Joelle van der Walt
*Duke University Center for Human
Genetics*
Genotype and Phenotype

Jeffery M. Vance
*Duke University Center for Human
Genetics*

Gene Discovery
Genomic Medicine
Genotype and Phenotype
Sanger, Fred

Gail Vance
Indiana University
Chromosomal Banding

Jeffrey T. Villinski
*University of Texas/MD Anderson
Cancer Center*
Sex Determination

Sue Wallace
Santa Rosa, California
Hemoglobinopathies

Giles Watts
Children's Hospital Boston
Cancer
Tumor Suppressor Genes

Kirk Wilhelmsen
*Ernest Gallo Clinic & Research
Center*
Addiction

Michelle P. Winn
Duke University Medical Center
Physician Scientist

Chantelle Wolpert
*Duke University Center for Human
Genetics*
Genetic Counseling
Genetic Discrimination
Nomenclature
Population Screening

Harry H. Wright
*University of South Carolina School
of Medicine*
Intelligence
Psychiatric Disorders
Sexual Orientation

Janice Zengel
University of Maryland, Baltimore
Ribosome
Translation

Stephan Zweifel
Carleton College
Mitochondrial Genome

Table of Contents

VOLUME 1

PREFACE	v
FOR YOUR REFERENCE	ix
LIST OF CONTRIBUTORS	xvii

A

Accelerated Aging: Progeria	1
Addiction	4
Aging and Life Span	6
Agricultural Biotechnology	9
Alternative Splicing	11
Alzheimer's Disease	14
Ames Test	19
Androgen Insensitivity Syndrome	21
Antibiotic Resistance	26
Antisense Nucleotides	29
Apoptosis	31
<i>Arabidopsis thaliana</i>	33
Archaea	36
Attention Deficit Hyperactivity Disorder	39
Attorney	42
Automated Sequencer	43

B

Balanced Polymorphism	45
Behavior	46
Bioinformatics	52
Biopesticides	57
Bioremediation	59
Biotechnology	62
Biotechnology Entrepreneur	65
Biotechnology: Ethical Issues	66
Biotechnology and Genetic Engineering, History	70
Birth Defects	74

Blood Type	82
Blotting	86
Breast Cancer	89

C

Cancer	92
Carcinogens	97
Cardiovascular Disease	101
Cell Cycle	103
Cell, Eukaryotic	108
Centromere	114
Chaperones	116
Chromosomal Aberrations	119
Chromosomal Banding	125
Chromosomal Theory of Inheritance, History	129
Chromosome, Eukaryotic	132
Chromosome, Prokaryotic	139
Chromosomes, Artificial	144
Classical Hybrid Genetics	146
Clinical Geneticist	149
Cloning Genes	152
Cloning: Ethical Issues	158
Cloning Organisms	161
College Professor	165
Colon Cancer	166
Color Vision	170
Combinatorial Chemistry	173
Complex Traits	177
Computational Biologist	181
Conjugation	182
Conservation Biology: Genetic Approaches	186
Conservation Geneticist	190
Crick, Francis	192
Crossing Over	194

Cycle Sequencing	198
Cystic Fibrosis	199

D

Delbrück, Max	203
Development, Genetic Control of	204
Diabetes	209
Disease, Genetics of	213
DNA	215
DNA Footprinting	220
DNA Libraries	222
DNA Microarrays	225
DNA Polymerases	230
DNA Profiling	233
DNA Repair	239
DNA Structure and Function, History	248
DNA Vaccines	253
Down Syndrome	256
PHOTO CREDITS	259
GLOSSARY	263
TOPICAL OUTLINE	281
INDEX	287

VOLUME 2

FOR YOUR REFERENCE	v
LIST OF CONTRIBUTORS	xiii

E

Educator	1
Embryonic Stem Cells	3
Epidemiologist	6
Epistasis	7
<i>Escherichia coli</i> (<i>E. coli</i> bacterium)	9
Eubacteria	11
Eugenics	16
Evolution, Molecular	21
Evolution of Genes	26
Eye Color	31

F

Fertilization	33
Founder Effect	36
Fragile X Syndrome	39
Fruit Fly: <i>Drosophila</i>	42

G

Gel Electrophoresis	45
---------------------------	----

Gene	50
Gene and Environment	54
Gene Discovery	57
Gene Expression: Overview of Control ..	61
Gene Families	67
Gene Flow	70
Gene Targeting	71
Gene Therapy	74
Gene Therapy: Ethical Issues	80
Genetic Code	83
Genetic Counseling	87
Genetic Counselor	91
Genetic Discrimination	92
Genetic Drift	94
Genetic Testing	96
Genetic Testing: Ethical Issues	101
Genetically Modified Foods	106
Geneticist	110
Genetics	111
Genome	112
Genomic Medicine	118
Genomics	120
Genomics Industry	123
Genotype and Phenotype	125
Growth Disorders	129

H

Hardy-Weinberg Equilibrium	133
Hemoglobinopathies	136
Hemophilia	141
Heterozygote Advantage	146
High-Throughput Screening	149
HIV	150
Homology	156
Hormonal Regulation	158
HPLC: High-Performance Liquid Chromatography	165
Human Disease Genes, Identification of ..	167
Human Genome Project	171
Human Immunodeficiency Virus	178
Huntington's Disease	178
Hybrid Superiority	178

I

Immune System Genetics	178
Imprinting	183
<i>In situ</i> Hybridization	186
Inbreeding	189

Individual Genetic Variation	191
Information Systems Manager	192
Inheritance, Extranuclear	194
Inheritance Patterns	199
Intelligence	207
Internet	211
PHOTO CREDITS	215
GLOSSARY	219
TOPICAL OUTLINE	237
INDEX	243

VOLUME 3

FOR YOUR REFERENCE	v
LIST OF CONTRIBUTORS	xiii

L

Laboratory Technician	1
Legal Issues	3
Linkage and Recombination	4

M

Maize	8
Mapping	11
Marker Systems	15
Mass Spectrometry	18
McClintock, Barbara	21
McKusick, Victor	22
Meiosis	24
Mendel, Gregor	30
Mendelian Genetics	32
Metabolic Disease	37
Methylation	46
Microbiologist	50
Mitochondrial Diseases	51
Mitochondrial Genome	55
Mitosis	57
Model Organisms	60
Molecular Anthropology	62
Molecular Biologist	70
Morgan, Thomas Hunt	72
Mosaicism	76
Muller, Hermann	80
Multiple Alleles	82
Muscular Dystrophy	83
Mutagen	87
Mutagenesis	89

Mutation	93
Mutation Rate	98

N

Nature of the Gene, History	101
Nomenclature	106
Nondisjunction	108
Nucleases	112
Nucleotide	115
Nucleus	119

O

Oncogenes	127
Operon	131
Overlapping Genes	135

P

Patenting Genes	136
Pedigree	138
Pharmaceutical Scientist	142
Pharmacogenetics and Pharmacogenomics	144
Physician Scientist	147
Plant Genetic Engineer	149
Plasmid	150
Pleiotropy	153
Polymerase Chain Reaction	154
Polymorphisms	159
Polyploidy	163
Population Bottleneck	167
Population Genetics	171
Population Screening	175
Post-translational Control	178
Prenatal Diagnosis	182
Prion	187
Privacy	190
Probability	193
Protein Sequencing	196
Proteins	198
Proteomics	205
Pseudogenes	209
Psychiatric Disorders	213
Public Health, Genetic Techniques in ..	216
Purification of DNA	220
PHOTO CREDITS	223
GLOSSARY	227
TOPICAL OUTLINE	245
INDEX	251

VOLUME 4

FOR YOUR REFERENCE	v
LIST OF CONTRIBUTORS	xiii

Q

Quantitative Traits	1
---------------------------	---

R

Reading Frame	4
Recombinant DNA	5
Repetitive DNA Sequences	7
Replication	12
Reproductive Technology	19
Reproductive Technology: Ethical Issues ..	26
Restriction Enzymes	31
Retrovirus	34
Reverse Transcriptase	39
Ribosome	42
Ribozyme	44
RNA	46
RNA Interference	54
RNA Processing	57
Rodent Models	60
Roundworm: <i>Caenorhabditis elegans</i>	62

S

Sanger, Fred	64
Science Writer	65
Selection	67
Sequencing DNA	69
Severe Combined Immune Deficiency ...	74
Sex Determination	78
Sexual Orientation	83
Signal Transduction	85
Speciation	91
Statistical Geneticist	93
Statistics	95

T

Tay-Sachs Disease	98
Technical Writer	102
Telomere	104
Transcription	106
Transcription Factors	112
Transduction	117
Transformation	121
Transgenic Animals	124
Transgenic Microorganisms	127
Transgenic Organisms: Ethical Issues ..	129
Transgenic Plants	132
Translation	135
Transplantation	139
Transposable Genetic Elements	143
Triplet Repeat Disease	148
Tumor Suppressor Genes	153
Twins	155

V

Viroids and Virusoids	162
Virus	164

W

Watson, James	171
---------------------	-----

X

X Chromosome	173
--------------------	-----

Y

Y Chromosome	176
Yeast	179

Z

Zebrafish	181
-----------------	-----

PHOTO CREDITS	185
---------------------	-----

GLOSSARY	189
----------------	-----

TOPICAL OUTLINE	207
-----------------------	-----

CUMULATIVE INDEX	213
------------------------	-----